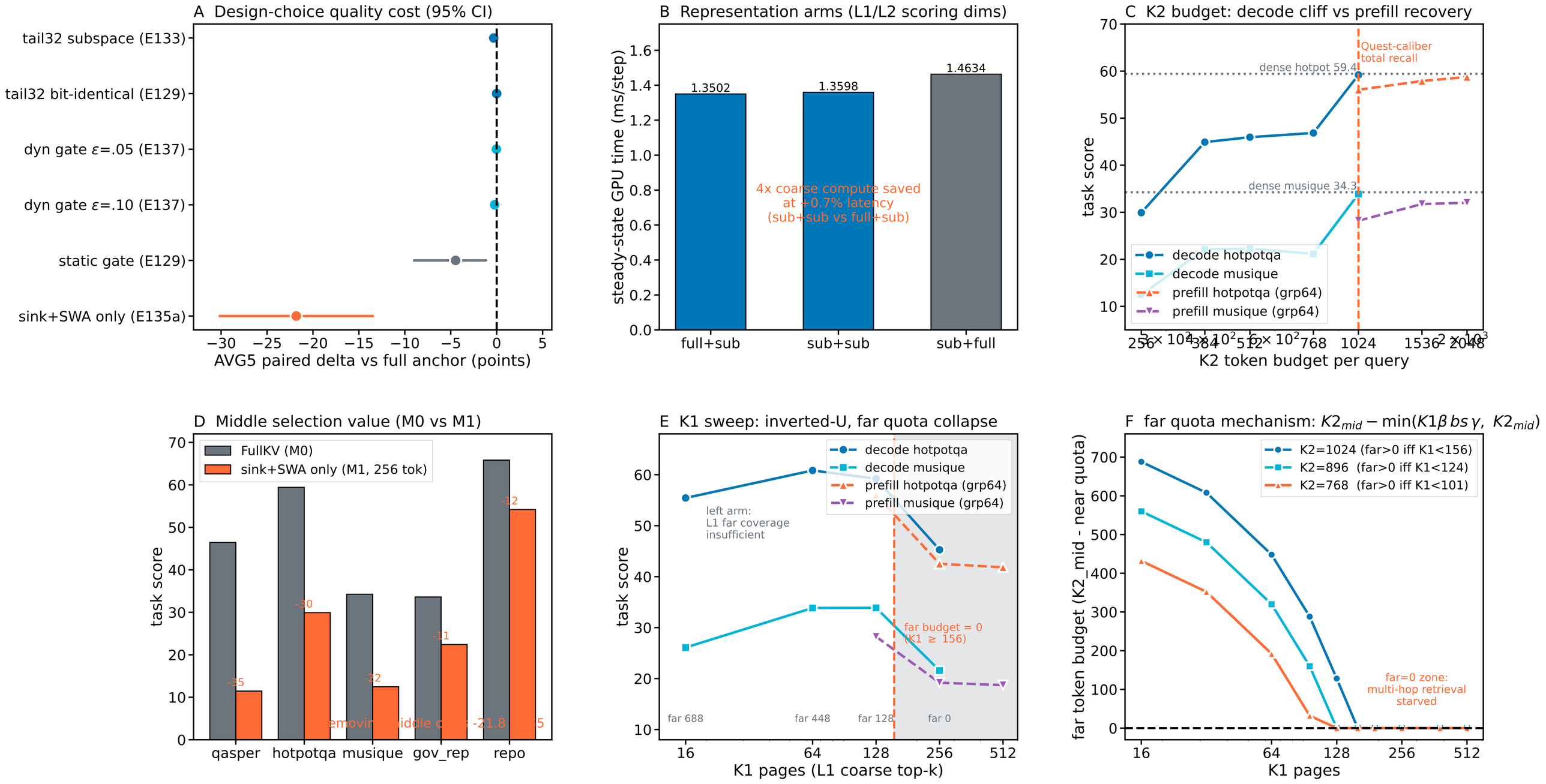
A Design-choice quality cost (95% CI)
tail32 subspace (E133)
tail32 bit-identical (E129)
dyn gate ε=.05 (E137)
dyn gate ε=.10 (E137)
static gate (E129)
sink+SWA only (E135a)
−30 −25 −20 −15 −10 −5 0 5
AVG5 paired delta vs full anchor (points)
B Representation arms (L1/L2 scoring dims)
steady-state GPU time (ms/step)
1.3502
1.3598
1.4634
4x coarse compute saved
at +0.7% latency
(sub+sub vs full+sub)
full+sub
sub+sub
sub+full
C K2 budget: decode cliff vs prefill recovery
task score
Quest-caliber
total recall
dense hotpot 59.4
dense musique 34.3
decode hotpotqa
decode musique
prefill hotpotqa (grp64)
prefill musique (grp64)
K2 token budget per query
D Middle selection value (M0 vs M1)
FullKV (M0)
sink+SWA only (M1, 256 tok)
task score
-35
-30
-22
-11
-12
qasper
hotpotqa
musique
gov_rep
repo
E K1 sweep: inverted-U, far quota collapse
decode hotpotqa
decode musique
prefill hotpotqa (grp64)
prefill musique (grp64)
left arm:
L1 far coverage
insufficient
far budget = 0
(K1 ≥ 156)
far 688
far 448
far 128
far 0
task score
K1 pages (L1 coarse top-k)
F far quota mechanism: K2_mid − min(K1β bs γ, K2_mid)
K2=1024 (far>0 iff K1<156)
K2=896 (far>0 iff K1<124)
K2=768 (far>0 iff K1<101)
far=0 zone:
multi-hop retrieval
starved
far token budget (K2_mid - near quota)
K1 pages